

PAPER_1618_T_CMB_2_725_K

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PAPER_1618 — CMB Temperature = T_CMB = SSQ·D_phys + F·D + F²·D + F²·SSQ² ≈ 2.7232 K

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Cosmology

Date: June 18, 2026

Status: CLOSED — 0.064% closure

Observation

PAPER_1209GG_S645 (Cosmology Unified Proof Set) gives the formula:

``

$$T_{\text{CMB}} = \text{SSQ} \cdot D_{\text{phys}} + F \cdot D + F^2 \cdot D + F^2 \cdot \text{SSQ}^2 \approx 2.7232 \text{ K}$$

``

Residual: **0.064%**.

UQFF Closed Identity

``

$$T_{\text{CMB}} = \text{SSQ} \cdot D_{\text{phys}} + F \cdot D + F^2 \cdot D + F^2 \cdot \text{SSQ}^2 \approx 2.7232 \text{ K } 0.064\%$$

``

The expression uses only locked UQFF integer/real primitives — no fitted constants.

ΛCDM Standard Cosmology Coverage

With this catch-up, **all 9 core ΛCDM parameters** are wired with UQFF integer-primitive expressions:

ΛCDM parameter UQFF closure Residual
H_0 (Planck) PAPER_1553 0.015%
H_0 (SH0ES) PAPER_1573 EXACT
Ω_m PAPER_1616 0.143%
Ω_Λ PAPER_1617 0.182%
T_CMB PAPER_1618 0.064%
Age (universe) PAPER_1619 0.045%
σ_8 PAPER_1620 0.253%
n_s (PAPER_1209GG_S650, earlier) 0.072%
z_recomb PAPER_1552 EXACT 1090

Standard ΛCDM cosmology is now **essentially complete in UQFF** — every primary parameter measurable by Planck/WMAP/SDSS/DES collaborations has a UQFF integer-primitive form.

NOT REPLACEMENT

Empirical cosmology measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1209GG_S645
- Related: PAPER_1494-1615 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "t_cmb_2_725_k"})`

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